Aravind Pradeep

AI Engineer | Agentic Systems & RAG Pipelines | LLMOps & Evaluation

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PROFESSIONAL SUMMARY

AI Engineer with two years at Perinet building the backend of an industrial-IoT platform: Python and Go services over MQTT streams, FastAPI, Docker, Kubernetes, and CI/CD, plus the model-benchmarking and marketing-analytics workstreams. Independently I design, ship, and maintain RAG, agent, and LLMOps systems: hybrid retrieval (BM25 + dense with reciprocal rank fusion) with RAGAs and MLflow evaluation, LangGraph and GraphRAG agent pipelines, QLoRA fine-tuning, and call-level observability. First author of arXiv:2607.02612 (DSD 2026). Nearly three years of professional software experience in total, including earlier enterprise work at Cognizant; M.Sc. in AI at BTU Cottbus, thesis phase.

PROFESSIONAL EXPERIENCE

Independent AI Engineering & M.Sc. Thesis Research

Open source (github.com/axon011) / BTU Cottbus | Berlin, Germany | Jun 2026 – present

- First-author paper on adaptive ViT token compression accepted at DSD 2026 (arXiv:2607.02612), thesis in progress; shipping the open-source RAG and agent portfolio below, including a live multi-agent German tutor.

AI Engineer

Perinet GmbH (20-person industrial-IoT company) | Cottbus, Germany | Jul 2024 – May 2026 (23 months)

- Developed Python and Go backend services connecting LLM workflows to live MQTT sensor streams, exposed as versioned FastAPI REST APIs with async endpoints; containerized with Docker/Kubernetes and automated CI/CD via GitHub Actions for zero-touch production releases.
- Led the containerization workstream for the AI Container umbrella project — PeriChat (Go-based chatbot with MapReduce retrieval), MessageSense (streaming MQTT message anomaly detection), and PeriXplore (sensor data exploration); sole maintainer of several of these services end to end, handling profiling, QA, and production deployment integration with the engine and ML teams.
- Owned the RAG evaluation and model-benchmarking workstream; built the hybrid retrieval and evaluation layer behind PeriChat and measured retrieval speed (sub-300ms p95), generation quality, and trade-offs across model variants for the AI container platform, informing downstream chatbot and corpus design.
- Built a marketing-analytics pipeline (CEO-sponsored) pulling LinkedIn and Google Analytics 4 data through MQTT into a dashboard, with an event-matching script that links campaigns to engagement; cut reporting lag from roughly 48 hours to a daily refresh.

Software Engineer Trainee

Cognizant Technology Solutions | India | Oct 2021 – May 2022

- Developed and maintained mainframe applications (COBOL, JCL, DB2) in agile sprints; built structured debugging practices and production deployment discipline across enterprise banking systems.

PUBLICATIONS

- **Fusion: A Framework for Unified Sequential Token Adaptation in Vision Transformers.** A. Pradeep, S. Nazari, M. Taheri, C. Herglotz. Accepted at DSD 2026. arXiv:2607.02612 [cs.CV], Jul 2026.
- Stages token merging, early exiting, and token pruning so they cooperate; on ImageNet-1k with DeiT-S it matches state-of-the-art adaptive ViT methods at comparable compute while cutting inference energy 48%.

PROJECTS

Deutsch-Tutor — Adaptive AI German Tutor | tutor.aravindpradee.me | Next.js, TypeScript, Tailwind, SSE Streaming, Zod, LLM Provider Seam

- Built a full-stack AI German tutor (Next.js App Router, TypeScript) that logs every grammar mistake from live conversation and generates personalized drills from them; five tabs span 32 CEFR-graded lessons (A1–B2), streaming chat, practice, a progress recommender, and a 40-rule grammar reference.
- Designed a two-agent-per-turn flow — the reply streams instantly over SSE while a parallel Corrector returns Zod-validated exact-substring error spans — behind a provider seam that swaps the LLM backend (Claude CLI locally, Gemini when deployed) via one env var; the practice recommender is ranked by an explainable recency-weighted formula, not an LLM.

RAG Evaluation System | FastAPI, Qdrant, LangChain, RAGAs, MLflow, BM25 | rag-eval-system

- Engineered a hybrid retrieval pipeline (dense embeddings + BM25 + Reciprocal Rank Fusion) over a 50-topic corpus using Qdrant, reaching 0.94 hit@5 and 0.96 citation presence on a 50-question evaluation set, with async embedding pre-computation and semantic caching.
- Built an automated evaluation pipeline measuring faithfulness, relevance, and context recall via RAGAs; tracked prompt and retrieval experiments in MLflow with regression alerts when retrieval quality dropped below baseline.

GraphRAG Agent | Python, networkx, LLM Entity/Relation Extraction, Docker | graphrag-agent

- Built a GraphRAG pipeline that constructs a typed, deduplicated knowledge graph from a document corpus via LLM entity/relation extraction (networkx), so retrieval follows explicit relationships instead of chunk-embedding similarity alone.

- Implemented k-hop subgraph retrieval producing grounded answers cited back to source, handling multi-document and relationship-spanning questions that flat RAG misses; LLM backend pluggable across Claude, Codex, and Gemini with a test suite.

Multi-Agent Research & Report Pipeline | LangGraph, CrewAI, FastAPI, Docker | multi-agent-pipeline

- Architected a 4-agent pipeline (Planner, Researcher, Writer, Critic) using LangGraph state machines with strict role boundaries and Pydantic structured outputs, producing 2,000+ word research reports with verifiable source citations.
- Deployed via async FastAPI with Docker containerization and GitHub Actions CI/CD, automating the path from code push to deployment with no manual steps.

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, CrewAI, Coding Agents (Claude Code, Codex), RAG, GraphRAG, Prompt Engineering, Structured Outputs (Pydantic), Tool Use, Function Calling, MCP

LLMOps & Evaluation: Langfuse (Tracing, Evals), RAGAs (Faithfulness, Relevance, Recall), MLflow, LLM-as-Judge, A/B Prompt Evaluation, Regression Detection, Guardrails, LLM Cost Tracking

ML & NLP: Python, PyTorch, Hugging Face Transformers, scikit-learn, Natural Language Processing (NLP), Generative AI (GenAI), Fine-Tuning (QLoRA, PEFT), Model Compression (Quantization, Pruning), Edge AI Optimization, Chunking, Embeddings, NER, Computer Vision, Vision Transformers (ViT), Explainable ML

LLMs & APIs: OpenAI, Anthropic (Claude), Gemini, DeepSeek, Open-Source LLMs (Qwen, DeepSeek, Llama)

Backend & Infra: Go, FastAPI, Data Pipelines, Data Integration, NoSQL Databases, Data Visualization, Automated Testing, REST/gRPC, Microservices, Distributed Systems, API Integrations, Docker, Kubernetes, Container Platforms, Linux, Vector Databases, GitHub Actions, GitLab CI/CD, SQL, PostgreSQL, Qdrant, ChromaDB, OpenSearch, Elasticsearch, Azure, AWS (S3, Bedrock, EC2)

Frontend: JavaScript, TypeScript, React, Next.js, Recharts

Languages: English (C1), German (B1), Malayalam (Native)

EDUCATION

M.Sc. Artificial Intelligence (Research Profile)

Brandenburg University of Technology | Cottbus, Germany | Oct 2022 – present (thesis phase)

- Thesis: Content-Aware ViT Optimization on Edge Devices (Computer Vision, Explainable ML).

B.Sc. Computer Application

BVM Holy Cross College | Kottayam, India | Jul 2018 – Mar 2021